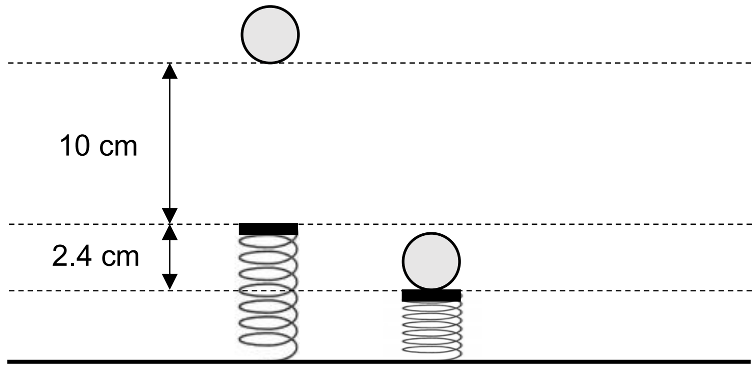
12 A 20 g ball bearing is released from rest 10 cm above the top of an unstretched spring. It compresses the spring and comes to rest when the spring is compressed by 2.4 cm as shown in the figure below.



What is the spring constant of the spring?

A 2.0 N m^{-1} **B** 8.6 N m^{-1} **C** 68 N m^{-1} **D** 84 N m^{-1}

Ans: D

Loss in GPE = gain in elastic potential energy

$$mg(h+x)=\frac{1}{2}kx^2$$

$$(0.020)(9.81)(0.10+0.024)=\frac{1}{2}k(0.024)^2$$

$$k=84 \text{ N m}^{-1}$$